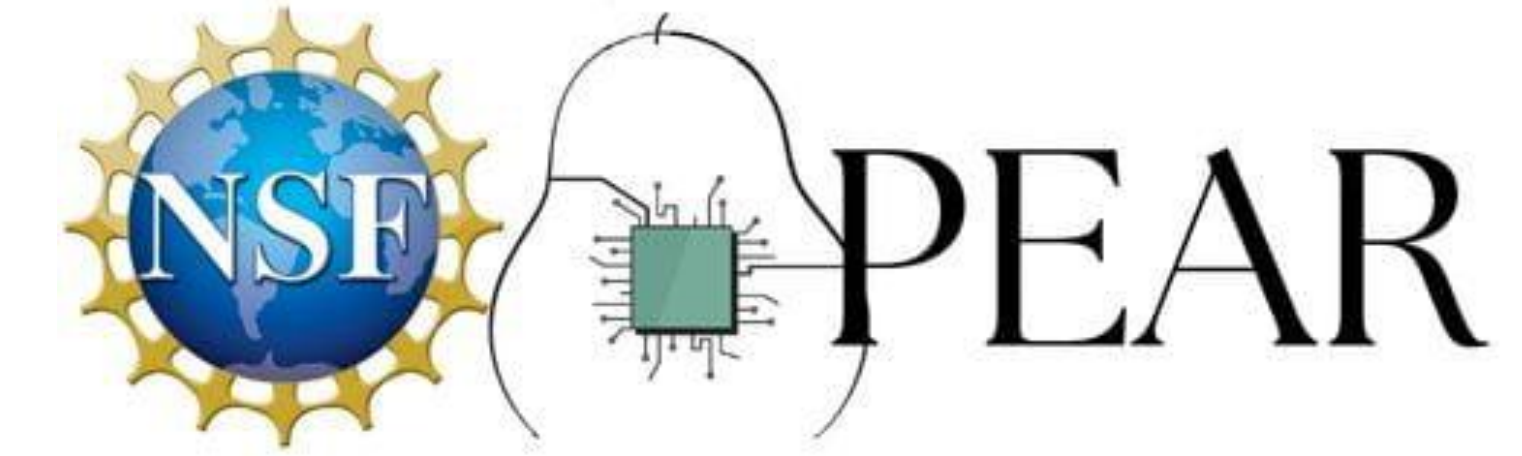




Why Can Large Language Models Optimize?



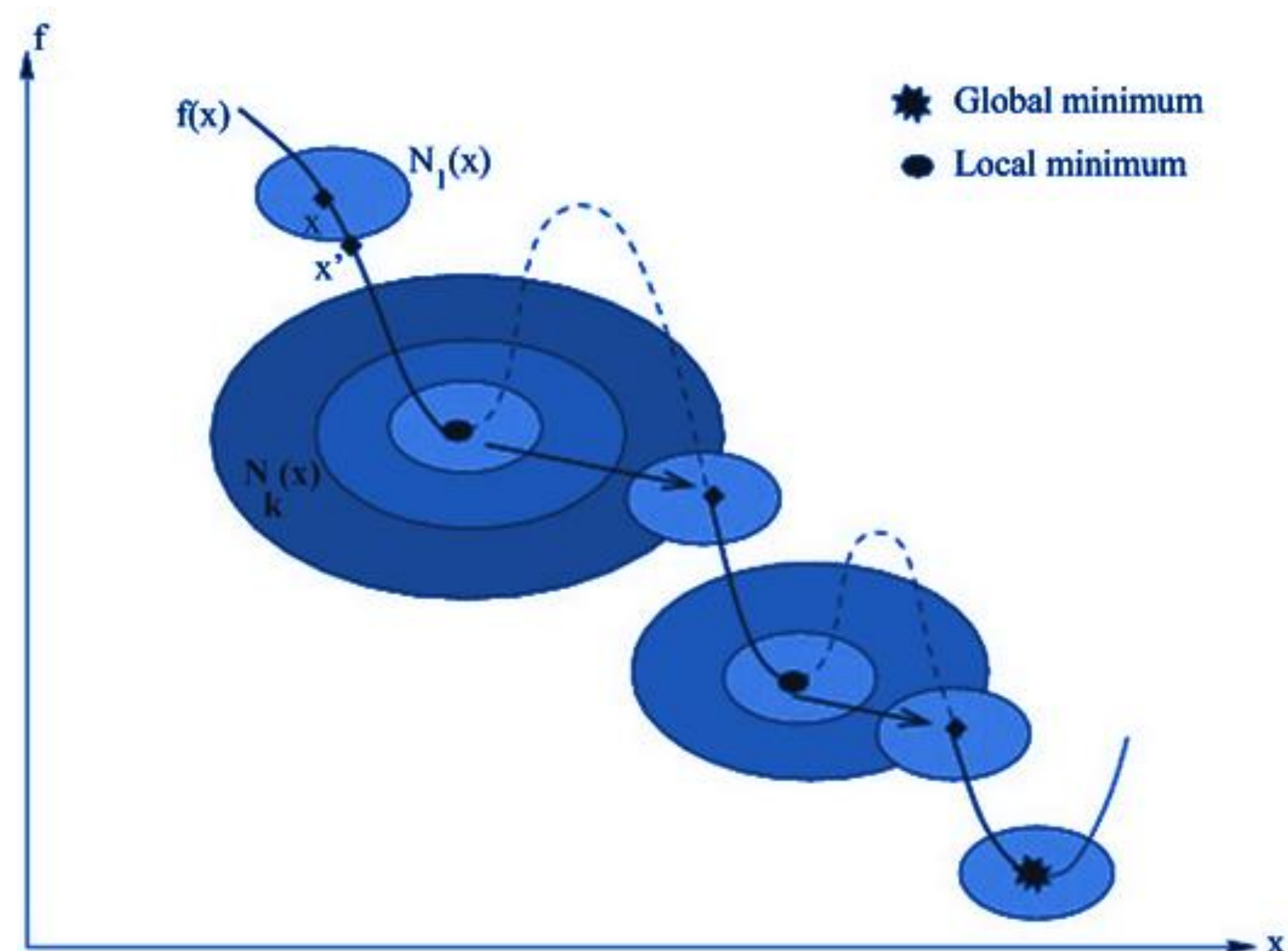
Aidan Riordan Riordan.a@northeastern.edu Advisor: Ozlem Ergun

Variable Neighborhood Search (VNS)

As our world becomes more interconnected, there is an ever-increasing demand to solve larger more complex problems.

Metaheuristics are used to solve these optimization problems such as VNS, Genetic Algorithms, and Particle Swarm Optimization.

Why VNS - it is more **interpretable**, easy to use and fast to learn. Can be **adapted** to solve wide range of problems. And integrated within other optimization / machine learning approaches at **scale**.



VNS explores small, local areas while a solution is improving, but systematically jumps to larger search areas when stuck in a local optimum.

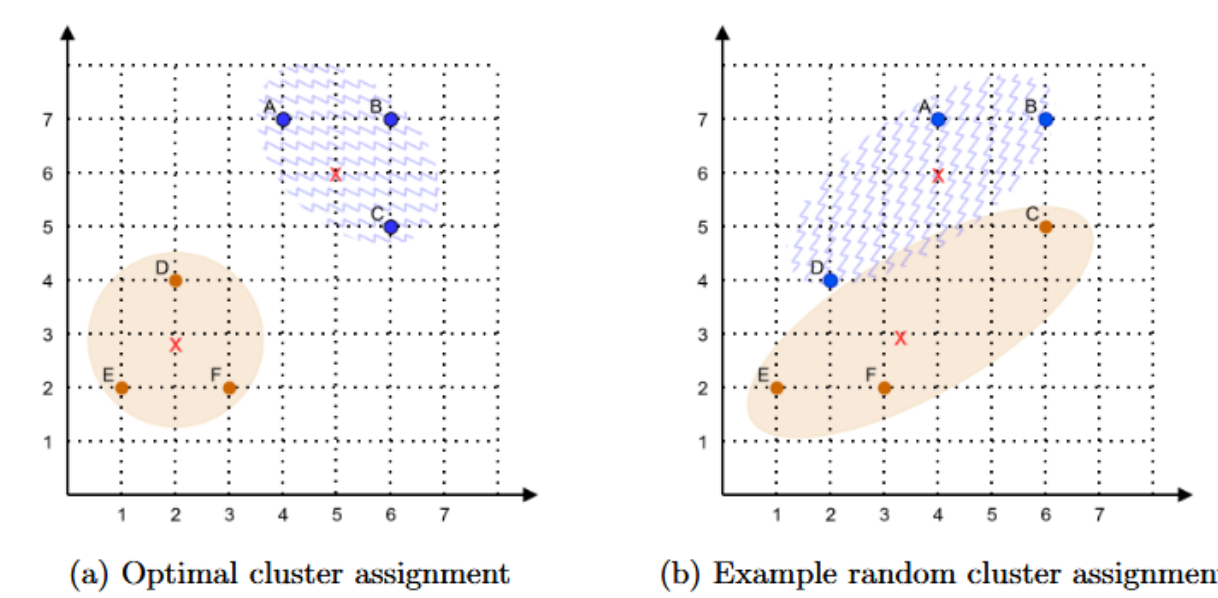
Scan for Paper



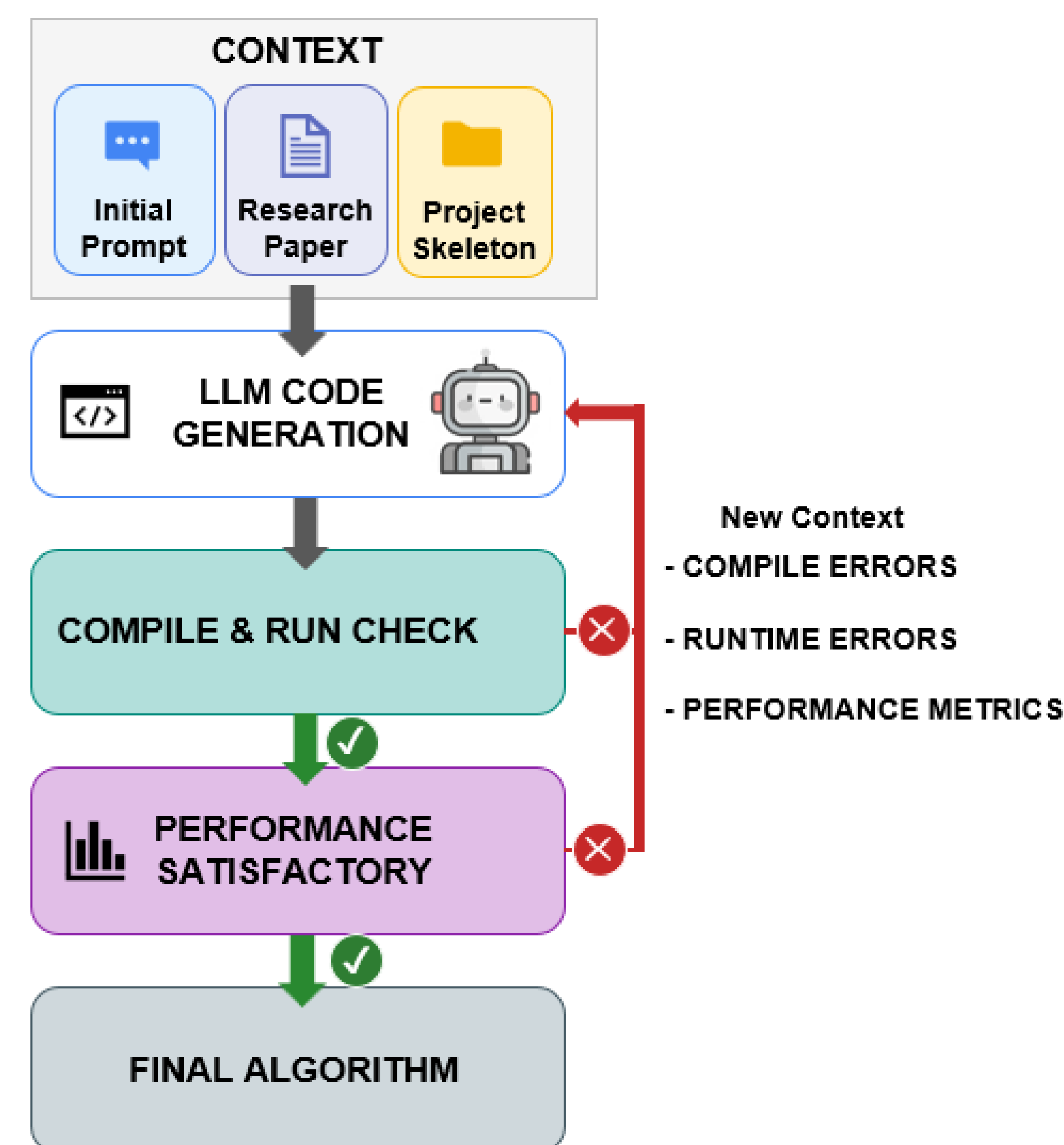
Interpretability, Adaptability and Scalability of Variable Neighborhood Search

Can LLMs do VNS?

We give LLMs a research paper describing a VNS algorithm for a clustering problem and test if it can make code matching the original algorithm's performance (original code not publicly available or given to LLM).



Balanced clustering problem – assign groups to data ensuring equal group size and minimizing distance from center and each point.



✓ **Gemini 2.5 Pro** successfully replicated the original algorithm's performance ($p < .001$) and corrected for an error in the original paper.

✗ **Claude Sonnet 4** replicated the error in the paper and underperformed the original paper's algorithm.

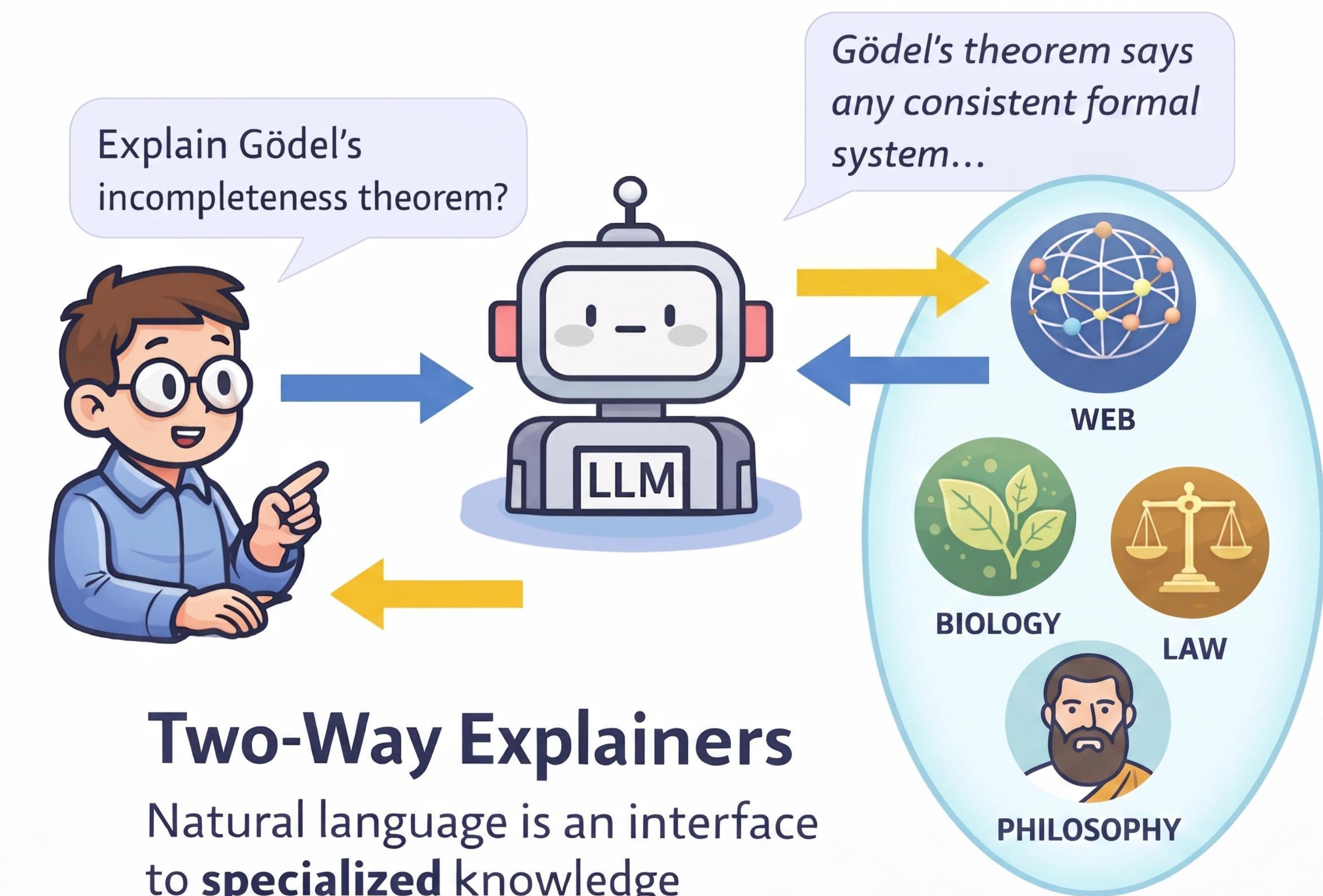
Large Language Models for Metaheuristic Implementation: A Case Study with Variable Neighborhood Search

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Why can LLMs do this?

What enables LLMs to generalize and be useful to new problems in different specialized domains? We propose a framework to understand LLMs, and how to best leverage them.



Two-Way Explainers
Natural language is an interface to **specialized** knowledge

Award-Winning Digital Subject-Matter Expert (NAM 2025, Jason Hodges et al)
In manufacturing, sensor data and alarm codes are "black boxes" to non-specialists. DSME is an LLM powered application that interprets sensor code in real-time to provide machine-level insights.



LLMs connect the near-entirety of digitized human knowledge into a single embedding space – building features and concepts from it. Their capability relies on the advanced retrieval of these existing concepts, allowing us to extract insights from highly specialized disciplines.

The Kaleidoscope of Knowledge and the Iron Man Suit: LLMs are Large Knowledge Models that Function as Two-Way Explainers for the Complex Systems Humans Evolved with Language

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